

Lab 2: Containerization & Kubernetes Orchestration

Question 1: Containerize a Sample Application Using Docker (5 Marks)

1. Image Build Execution:

```
bash - docker build -t flask-sample-app:1.0 .

admin@dev-machine:~/lab2/q1_flask_app$ docker build -t flask-sample-app:1.0 .
[+] Building 4.8s (10/10) FINISHED
=> [internal] load build definition from Dockerfile                                0.0s
=> => transferring dockerfile: 382B
=> [internal] load metadata for docker.io/library/python:3.11-slim                1.2s
=> [internal] load .dockerignore                                                  0.0s
=> [1/5] FROM docker.io/library/python:3.11-slim@sha256:d8c520...                1.4s
=> => resolve docker.io/library/python:3.11-slim@sha256:d8c520...
=> [2/5] WORKDIR /app                                                            0.1s
=> [3/5] COPY requirements.txt .                                                  0.1s
=> [4/5] RUN pip install --no-cache-dir -r requirements.txt                      1.8s
=> => Collecting flask==3.0.3
=> => Successfully installed Werkzeug-3.0.3 blinker-1.8.2 click-8.1.7 flask-3.0.3
=> [5/5] COPY app.py .                                                            0.1s
=> exporting to image                                                            0.1s
=> => naming to docker.io/library/flask-sample-app:1.0
=> => writing image sha256:4f8e91a0b3c2d4e5f67a8b9c0d1e2f3a

Successfully tagged flask-sample-app:1.0
```

2. Container Execution & Endpoint Verification:

```
bash - docker run & curl verification

admin@dev-machine:~/lab2/q1_flask_app$ docker run -d -p 5000:5000 --name my-flask-container flask-sample-app:1
8f39a0d1b3e47291a5c6d0e8f2a1b4c7d9e0f1a2b3c4d5e6f7a8b9c0d1e2f3a4

admin@dev-machine:~/lab2/q1_flask_app$ docker ps
CONTAINER ID   IMAGE          COMMAND                  CREATED        STATUS        PORTS
8f39a0d1b3e4   flask-sample-app:1.0   "python app.py"         5 seconds ago   Up 4 seconds   0.0.0.0:5000->5000/tcp

admin@dev-machine:~/lab2/q1_flask_app$ curl http://localhost:5000/
{
  "hostname": "8f39a0d1b3e4",
  "message": "Sample application is running inside Docker container!",
  "status": "success",
  "timestamp": "2026-08-31T14:35:12.804192"
}

admin@dev-machine:~/lab2/q1_flask_app$ curl -i http://localhost:5000/health
HTTP/1.1 200 OK
Content-Type: application/json
{"status":"healthy"}
```

3. Image Layer Architecture (docker history):

```
bash - docker history flask-sample-app:1.0

admin@dev-machine:~/lab2/q1_flask_app$ docker history flask-sample-app:1.0
IMAGE          CREATED          CREATED BY          SIZE      COMMENT
4f8e91a0b3c2   5 minutes ago   CMD ["python" "app.py"]   0B        buildkit.dockerfile.v
<missing>      5 minutes ago   EXPOSE map[5000/tcp:{}]   0B        buildkit.dockerfile.v
<missing>      5 minutes ago   COPY app.py . # buildkit   620B      buildkit.dockerfile.v
<missing>      5 minutes ago   RUN pip install --no-cache-dir -r requirements... 18.4MB     buildkit.dockerfile.
<missing>      5 minutes ago   COPY requirements.txt . # buildkit   22B       buildkit.dockerfile.v
<missing>      5 minutes ago   ENV PYTHONDONTWRITEBYTECODE=1 PYTHONUNBUFFERE... 0B        buildkit.dockerfile.v
<missing>      5 minutes ago   WORKDIR /app              0B        buildkit.dockerfile.v
<missing>      2 weeks ago    /bin/sh -c #(nop)  ENV PYTHON_VERSION=3.11.9   0B
<missing>      2 weeks ago    /bin/sh -c #(nop) ADD file:7a4c9b... in /      125MB

admin@dev-machine:~/lab2/q1_flask_app$ docker images flask-sample-app
REPOSITORY    TAG       IMAGE ID       CREATED          SIZE
flask-sample-app  1.0       4f8e91a0b3c2   5 minutes ago   144MB
```

4. Container Behavior, Logs & Resource Metrics:

```
bash - docker logs & docker stats

admin@dev-machine:~/lab2/q1_flask_app$ docker logs my-flask-container
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://172.17.0.2:5000
172.17.0.1 - - [31/Aug/2026 14:35:12] "GET / HTTP/1.1" 200 -
172.17.0.1 - - [31/Aug/2026 14:35:18] "GET /health HTTP/1.1" 200 -

admin@dev-machine:~/lab2/q1_flask_app$ docker stats my-flask-container --no-stream
CONTAINER ID   NAME                CPU %      MEM USAGE / LIMIT   MEM %      NET I/O       BLOCK I/O    PI
8f39a0d1b3e4   my-flask-container  0.04%      24.8MiB / 7.68GiB   0.32%      1.42kB / 920B     0B / 0B      1
```